

ASSIGNMENT – RISC-V SOFTCORE IMPLEMENTATION IN VERILOG

Requirements

You are required to implement a single cycle data path according to the RV32I ISA in Verilog. You may refer to figures in Computer Organization and Design¹. You must implement the following instructions: add, sub, addi, lw, sw, beq, jal.

You may reuse any code that you have written in tutorials or your portfolio. You may use any Verilog features (e.g. you can use the + operator rather than build a ripple carry adder). Ensure your code is well documented with comments and is easy to understand.

You must write a test programme in RISC-V assembly, assemble it to machine language in hexadecimal format (you may do this in RARS or any RISC-V assembler of your choice). You must write a test bench file into which you will load and run your test RISC-V assembly programme.

Submission

You must submit a single report file in PDF format to the assignment drop box in MyLO. The report should include the following sections:

- **Introduction:** Introduce the project and summarise the report. 250 words or less.
- **Technical Detail:** Provide a brief technical explanation to address criteria 2 and 3. Do not include code listings here.
- **Testing:** Briefly explain how you tested your implementation in terms of addressing all requirements. Refer to criterion 4
- **Conclusion:** Include one paragraph to summarise the report. Optionally you may add a paragraph to mention any other details or issues that were not addressed in the body of the report.
- **Code Listings:** Include all code files. Each file must have its own heading. Code must be formatted neatly and preferably be syntax highlighted². Do not use screenshots for code listings. Refer to criteria 1 and 4.

¹ The text book we are using for this unit is the RISC-V edition which is available in the Morris Miller library. You can also access the MIPS 4th edition electronically in the UTas library. MIPS is a very similar ISA to RISC-V and most of the components are the same.

² You can copy your code into Visual Studio Code (or any similar IDE) and then copy from the IDE into Word to get syntax highlighting.

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Assessment

The following criteria will be assessed in this assignment:

Criterion	Description	Learning Outcome
1	Simulate an advanced digital circuit.	ILO2
2	Calculate performance of advanced digital circuit.	ILO2
3	Present a suitable design to meet requirements.	ILO5
4	Evaluate the design to ensure it meets the requirements.	ILO5

Criterion 1 (30%): Simulate an advanced digital circuit.

For full marks you must demonstrate the following through your report. Marks will be deducted for each item not adequately demonstrated.

1. Provide full code listings.
2. Meet requirements through coding Verilog with correct logic.
3. Adequately document your code with comments.
4. Use appropriate datatypes (reg, wire, bit-widths, arrays).
5. Make appropriate use of constants.
6. Structure your code appropriately for ease of reading.

Criterion 2 (10%): Calculate performance of an advanced digital circuit.

For full marks you must demonstrate the following through your report. Marks will be deducted for each item not adequately demonstrated.

1. Present the usage of FPGA internal components
2. Comment on the performance requirements of your design (e.g. clock speed limitations, size requirements) and the suitability of the FPGA used in the Basys3 development boards. You may need to do some research to complete this point.

Criterion 3 (30%): Present a suitable design to meet requirements.

For full marks you must demonstrate the following through your report. Marks will be deducted for each item not adequately demonstrated.

1. Present a block diagram to illustrate the full set of modules and their interactions and dependencies.
2. Explain the hierarchy of modules, how they work together and interact.
3. Provide a correct description of all modules in the design. Describe their inputs, outputs, types and functionality.

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Criterion 4 (30%): Evaluate the design to ensure it meets the requirements.

For full marks you must demonstrate the following through your report. Marks will be deducted for each item not adequately demonstrated.

1. Develop a test programme in RISC-V assembly that tests a range of values for all required instructions.
2. Explain how the test programme adequately tests for all requirements.
3. Present successful results from running the test programme via the given test bench (use your own RISC-V assembly programme from point 1 above).